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AI-SMLA: An Artificial Intelligence based Smart Machine Learning Algorithm for Complex Image Segmentation Issues in Vertex Image Processing

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Abstract—**Today's machine learning is considered as one of the artificial intelligence technologies used in many ways. Its functions are very accurate, from receiving the given input data to calculating, measuring and outputting it. It is used to calculate a valid result without user intervention. In that order the applications of this machine learning method in digital image processing technology are constantly increasing. This technology is used for everything from overcoming facial identities to calculating the various photo identities of mobile devices currently in use. In this paper a smart machine learning based algorithm has been designed to effectively handle vertex image processing environment and complex image segmentation functions. Its special feature is that it examines the given photos based on its dead pixels and calculates the correct results. There will be no human intervention in this system. The proposed smart algorithm was compared with some existing algorithms.**

Keywords: Artificial intelligence, accurate, digital image processing, photo identities, mobile devices, machine learning, vertex image processing, image segmentation.

I. INTRODUCTION

The special features of the machine learning system in general are an arbitrary artificial intelligence system designed for individual use methods and to define the advanced process modules of organizations [4]. The speed of data handling and the ability to calculate efficient results in less time are the reasons why most people seek these. It is further enhanced by the accuracy of its results and its ability to make meaningful decisions. Image processing methods in general have come to a standstill today. The dominance of image processing technology has increased everywhere, from face scans to builder images on your smart phone cameras [8]-[11]. Furthermore the series of images generated by the image processing system has a number of functions that further enhance its accuracy. Each work package is involved in enhancing the data of specific images while executing it. Doing so tends to increase its size and the ratio of data such as resolution. Thus the impact on it will greatly increase with increasing work. For example, when an image is processed and its work is completed and the size of that particular image increases, it may need to be compressed to text it. Then when you compress them based on those image blocks it goes back to the old state. Current technologies help to rectify minor problems in image processing.Digital filters are used to blur and sharpen digital images [12]-[14]. Filtering in the location sphere can be done by rotating in the frequency field (Fourier) by covering specially designed cores (filter array) or certain frequency bands. Digital cameras usually have special hardware for image processing (special chips or circuits embedded in other chips) to convert source data from image sensors to a color-adjusted image in the form of a standard image file.

An image processing refers to the work that is done not only on the image but also on the enclosure attributes. Image processing is a combination of changes in different data sets such as brightness, sharpness, color, contrast, filter, resolution and size [15]-[16]. It is very helpful to analyze the information of those particular images and extract the required information. Due to the high level of development in recent times, some of the problems that arise with this technology are likely to be high [17]. At present most of the businesses are doing special activities like displaying their products, highlighting their advertisements from other prostitutes, impressing the customers through format authentication, and creating multiple displays in one frame through sorting processes [18]. All of these are based on the digital system [19]. Its operation is highly complex and its accuracy is high when there are some efficient and spontaneous machine learning methods to manage it [20]. The following volumes on this paper detail the information of previous scientists on it, the proposed algorithm, the experiment and the results.

II. RELATED WORKS

Abbas et al [1] introduced an improved image segmentation system. It carried out analyzes based on classified image blocks. The significance of this was that its various image blocks were interconnected by its link codes which were continuously divided into smaller wages based on its accuracy. The accuracy of those images was calculated based on these links. But only its coupling speed during grouping operations made connectivity more difficult.Bertozzi et al. [2] handled limited image sets to sound the map. The significance of this is that the image blocks divided into categories have its integration paths based on small groups. This method of converting it into a graph and configuring it based on its analytical results.Merkurjev et al [3] proposed a method of splitting the existing frame modules into video mode and making changes to its imaging mode. Image definitions give good results in this way but publishing them together as a video

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does not achieve excellent accuracy. Van Gennip et al [5] analyzed accurate maps defined on the basis of image volume. In this method the analysis and segmentation methods came out better and the image resolution that integrates it was better. But due to the large size of the images, problems arose in compressing it.Cavalcanti et al. [6] proposed microscopic image block classification. In this, special codes were given to the image divided into blocks based on the dot pixel and grouping functions were created based on it. This process gave good results.Chan.T.F et al [7] categorized image groups based on vectors. That is, each separated image block is linked to each vector and then its blocks are classified. But even though the accuracy of this muran was high, the time taken to link the images took a long time. And sometimes the images are incomplete vectors of connection where proper alignment of the pixels is not possible.

III. PROPOSED METHOD

The proposed Artificial intelligence based smart machine learning algorithm (AI-SMLA) is demonstrated in fig 1. A specific pipeline is always created when machine learning algorithms are created. Its main purpose is that the inputs given from that pipeline data take the learning steps required for machine learning methods. The currently proposed algorithm says the same thing. That is, the first proposed algorithm for machine learning should always use high-accuracy data to maximize its accuracy. Therefore its planning is designed to make it work well. High resolution should be ensured where images are obtained as input based on computer vision. This whole algorithm works on its basis. It can be used to add and remove high resolution images.

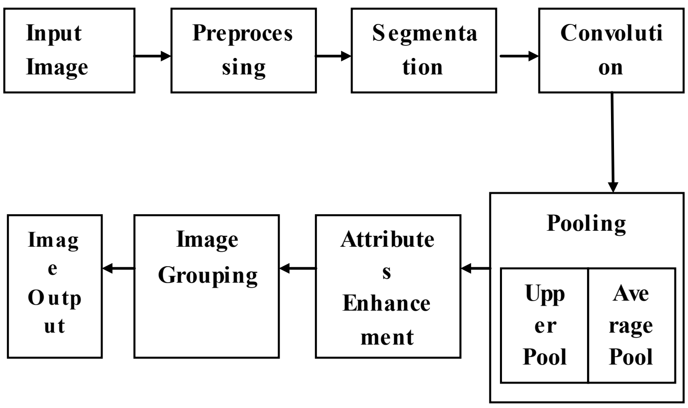

Fig 1: Proposed system block diagram

First all the inputs are extracted and then their entire format is changed to the extent required by the machine. The carving method is implemented by applying the required parts in the next given image. It does not know it as forces. Here the matrix processes present in the blocks of numbers form the basis of this carving process.

Algorithm : Artificial intelligence based smart machine learning algorithm

1. Start
2. Enter the input images
3. Slice the image as per the dot matrix method
4. Provide the input segment
5. If (segment = ordered)
6. Then convolute the image
7. Then pooling the image
8. Then enhance the attributes
9. Else go-to step 3
10. Group the images
17. end

That is, the input information first divides the blocks into pixel rows. Depending on the clarity of the given inputs divided into height * width * dimension.Based on this, the machine learning algorithm that calculates this selects the correct Measurement method. This helps to give known aspect vectors.The proposed algorithm consists of two modules.

3.1. CONVOLUTION LAYER:

It helps to handle complex image blocks. The matrix modules on which it is based help to identify the special features present in the image and extract the image module from the unwanted data. Image square blocks that are random in this layer set combine with the dot filter to further enhance its functionality. The higher the rating of the filter and patch modules the higher the output of the convolution module. Perhaps the amount of output available will be significantly lower when the performance of either of these two decreases. The proposed convolution layer operation is shown in figure 2.

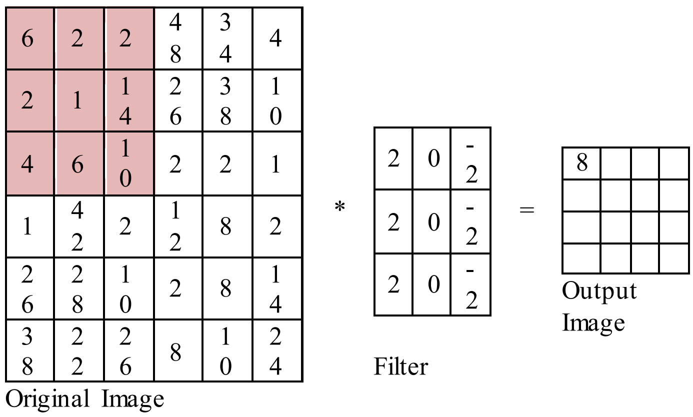

Fig 2: Operation of proposed convolution layer

The image segmentation up-to the 'n' th level image segments are denoted as follows,

$$IS\,(a) = \frac{\mu}{2}\int_{\sigma}^{0} |\nabla\, a(p)|^2\, dp + \frac{1}{\mu}\int_{\sigma}^{0} Q\,\big(a\,(p)\big)\, dp \quad (1)$$

Where,

IS (a) = convolution segment output;
σ = subnet of the given image
a(p) = two-phase image resolution concentration

3.2. POOLING LAYER:

When detecting certain features of images using the convolution layers mentioned above, those blocks will have multiple feature levels. These types of maps are created when the convolution process is used between the inserted image and the special filters. Thus the size of the image is likely to expand. This pooling process is used to reduce this. In this

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method the pixel values in each volume of the image are reduced to a maximum. These increase the accuracy of the workings of the machine learning system as they create movement individually for each set of inputs. The accuracy is further enhanced by the fact that their functions operate with two separate modules.The proposed pooling layer operation is shown in figure 3.

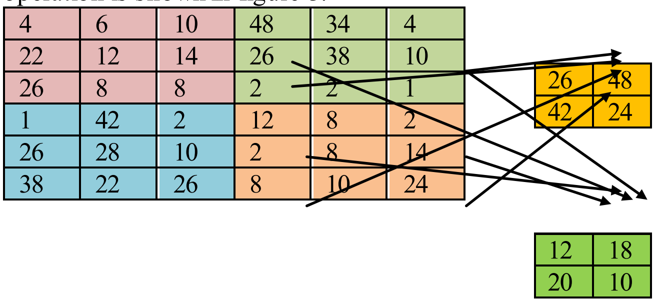

Fig 3: Operation of proposed pooling layer for the discrete operations,

$$a(p) = \sum_{b \in d} q\ (z, x) \qquad (2)$$

Where,

$q\ (z, x)$ = diagonal matrix of the given image

IV. Results and Discussion

The proposed Artificial intelligence based smart machine learning algorithm (AI-SMLA) was compared with the existing an improved segmentation method (ISM), The Diffuse interface models on graphs (DIMG), Graph MBO method for standoff detection (GMBO) and Macroscopic pigmented skin lesion segmentation (MPSLS)

There are the following parameters are evaluate the water quality. That is the Segmentation accuracy, input image recognition, input image rejection, Segmentation precision, Segmentation recall, and Segmentation F1-Score. Before understand the quality rate of the parameters, will know about the following,

Positive-T (TP)–It's the perfect predicted correct or above thecalibration level.

Negative-T (TN)–It's the negative prediction values below the calibration level.

Positives-F (FP) – When the exact values are in calibration level and the predicted samples are in same level

Negative-F (FN) – When the exact values are in calibration level butthe predicted samples are in different level

4.1. Measurementof Input image recognition

Usually the data given in the inputs are very large volumes. The basic data of the various image modules running on the basis of these modules must be largely formatted. Special features of this design will take into account the data from its various angles. The volumes of this data can be calculated as follows.

Then, the un-segmented image recognition of a method is given by

$$\textit{Input Image Recognition} = \sum_{a=1}^{h} Gj \qquad (3)$$

Where,Gj is the total number of segmented input image commands

The table 1 presents the Measurement of image recognition between existing ISM, DIMG, GMBO, MPSLSand proposed AISMLA

Table 1: Measurement of input image recognition

No.of Samples	Input image recognition in (%)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
100	75.18	81.51	80.15	79.27	94.75
200	75.51	83.01	80.74	81.14	95.76
300	76.85	84.12	81.72	81.97	95.92
400	77.99	84.5	82.93	82.88	96.88
500	79.04	85.51	84.07	83.8	96.45
600	79.75	86.44	85.18	85.13	97.65
700	81.05	87.44	85.88	86.21	97.81

4.2. Measurement Input image Rejection

The proposed algorithm usually separates the image blocks with different data given inside. Some pixels in that category have very low resolution. You can skip this and design other parts of that particular image gallery as it will take more time to upgrade. Thus its accuracy is not affected.

$$\text{Calculation of image rejection } x(t) = \frac{\text{skippedimages under the active segmentation } (x,t)}{\textit{Total segmented image groups under the process } (x,t)} \qquad (4)$$

The below table 2 presents the Measurement of picture block denial between existing ISM, DIMG, GMBO, MPSLSand proposed AISMLA

Table 2: Measurement of input image rejection

No.of Samples	Input image rejection in (%)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
100	24.82	18.49	19.85	20.73	5.25
200	24.49	16.99	19.26	18.86	4.24
300	23.15	15.88	18.28	18.03	4.08
400	22.01	15.5	17.07	17.12	3.12
500	20.96	14.49	15.93	16.2	3.55
600	20.25	13.56	14.82	14.87	2.35
700	18.95	12.56	14.12	13.79	2.19

4.3. Measurement of Segmentation accuracy:

The Segmentation accuracy is the parameter which describes the ratio between completely predicted Segmentation input images from the given pieces to the total number of collected image pieces. When the rate of Segmentation accuracy is high then the given output image sample getting high quality rate.

$$\text{Accuracy Measurement} = \frac{TP+TN}{\textit{All collected samples}} \qquad (5)$$

The below table 3demonstrates the various measurement comparison of the Segmentation accuracy values between the existing ISM, DIMG, GMBO, MPSLS and proposed AISMLA

4.4. Measurement of Segmentation precision:

Segmentation precision measurement is the ratio between the positive true samples and total true samples. The total true samples are calculated by the sum of positive true samples and false positive samples.

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Precision Measurement =

$$\frac{True\ Positive\ Predictions}{True\ Positive\ prediction + False\ Positive\ Prediction} \quad (6)$$

Table 3: Measurement of accuracy measurement

No.of Samples	Accuracy measurement in (%)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
100	72.88	79.21	83.55	82.01	93.84
200	73.21	80.71	84.14	83.88	94.88
300	74.55	81.82	85.12	84.71	95.01
400	75.69	82.2	86.33	85.62	95.97
500	76.74	83.21	87.47	86.54	95.54
600	77.45	84.14	88.58	87.87	96.78
700	78.75	85.14	89.28	88.74	96.89

The below table 4 demonstrates the various measurement comparison of the Segmentation precision values between the existing ISM, DIMG, GMBO, MPSLS and proposed AISMLA

Table 4: Measurement of precision measurement

No.of Samples	Precision measurement in (%)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
100	74.14	71.47	75.99	73.57	94.58
200	75.77	73.21	77.57	74.99	95.87
300	76.25	75.55	79.77	76.25	96.88
400	77.54	76.36	81.4	78.24	97.77
500	79.65	78.65	82.54	80.71	98.14
600	81.14	80.58	84.74	82.15	99.18
700	82.95	82.31	85.89	83.87	99.95

4.5. Measurement of Segmentation recall:

Segmentation recall measurement is the ratio between the positive true samples and the sum of positive true samples and false negative true samples.

Recall Measurement =

$$\frac{True\ Positive\ Predictions}{True\ Positive\ Predictions + False\ Negative\ Predictions} \quad (7)$$

The below table 5 demonstrates the various measurement comparison of the Segmentation recall values between the existing ISM, DIMG, GMBO, MPSLS and proposed AISMLA

Table 5: Measurement of recall rate

No.of Samples	Recall rate in (%)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
100	64.25	75.57	76.15	74.58	94.58
200	65.74	77.54	78.57	76.78	94.57
300	66.54	78.67	78.98	77.58	95.77
400	68.87	79.86	80.58	78.25	96.25
500	69.88	80.25	82.9	79.68	97.68
600	70.52	81.77	84.15	80.77	98.84
700	71.18	82.01	86.88	81.25	99.61

4.6. Measurement of Segmentation F1-Score:

It's measured by the average sample values of Segmentation precision and Segmentation recall of the samples.

$$\text{F1-Score Measurement} = \frac{2*(Recall*Precision)}{(Recall+Precision)} \quad (8)$$

The below table 6 demonstrates the various measurement comparison of the Segmentation F1-Score values between the existing ISM, DIMG, GMBO, MPSLS and proposed AISMLA

Table 6: Measurement of F1-Score

No.of Samples	F1-Score in (%)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
100	72.76	71.94	73.64	70.38	94.74
200	72.65	71.96	73.47	70.11	94.24
300	72.63	72.84	74.2	70.41	94.36
400	75.73	75.67	77.54	73.92	97.59
500	76.93	76.99	78.27	75.24	97.97
600	77.54	77.82	79.16	75.78	98.54
700	77.95	78.22	79.24	76.08	98.24

4.7. Measurement of Recognition duration:

The Measurement duration is nothing but the time taken to calculate the prediction of two different images.

$$\text{Recognition Duration} = \frac{No.of\ input\ samples}{Computaion\ Speed} \quad (9)$$

The below table 7 demonstrates the various measurement comparison of the Segmentation recognition duration between the existing ISM, DIMG, GMBO, MPSLS and proposed AISMLA

Table 7: Measurement of recognition duration

No.of Samples	Recognition Duration in (ms)				
	ISM	DIMG	GMBO	MPSLS	AISMLA
2000	3510	3844	2617	2771	1214
3000	4287	4401	3022	3155	1380
4000	5064	4958	3427	3539	1546
5000	5841	5515	3832	3923	1712
6000	6618	6072	4237	4307	1878
7000	7395	6629	4642	4691	2044
8000	8172	7186	5047	5075	2210

V. Conclusion

The Image classification is generally seen as a process implemented on a segmentation basis. But its accuracy can be overstated if the right choice is not made for the practical problems that arise in terms of segmentation. Sometimes its accuracy is due to its dot matrix classification. The current algorithm based on this has a high degree of accuracy (96.79%).Thence the proposed Artificial intelligence based

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smart machine learning algorithm (AI-SMLA) was provide better results while compared with the existing an improved segmentation method (ISM), The Diffuse interface models on graphs (DIMG), Graph MBO method for standoff detection (GMBO) and Macroscopic pigmented skin lesion segmentation (MPSLS). Thus the separation type can be felt to work better. And the process of connecting it is less time consuming and its merging method does not take much time. When this is practically possible its speed may vary depending on the speed of that hardware processor.

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